INCH-POUND

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MILITARY SPECIFICATION

INITIATOR, CARTRIDGE ACTUATED, M53 METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M53 Cartridge Actuated Initiator Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550	Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-413	Visual Inspection Guide for Elastomeric O-rings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 3730), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

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(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

NAVAL ORDNANCE STATION (CAGE Codes 19200 and 19203)

10520621	Initiator, Cartridge Actuated, M53 Metal Parts Assembly
8830994	Box, Fiberboard (Inner), Packing, Ammunition, For Initiator, Cartridge Actuated, M53
8830995	Box, Packing, Ammunition, For Initiator, Cartridge Actuated, M53
9217624	Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Initiator, Cartridge Actuated, M53

U.S. Air Force

TPO 00-731-9272	Initiator, Cartridge Actuated, M53

(Application for copies should be addressed to: Commanding Officer, Naval Ordnance Station, Data Control Branch (Code 3720), Indian Head, MD 20640-5035.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to the first article inspection (see 4.4 and 6.3).

3.2 Materials. All materials used in the manufacture of the initiators shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Raw material. The contractor shall be responsible for selecting from the classification list contained in the raw material specification the surface and physical condition of the raw material intended for use in manufacture.

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3.3 Hardness and other physical properties. Hardness and other physical properties required by the drawing shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.4 Hydrostatic pressure. The initiator chamber and cap shall withstand a hydrostatic pressure of 14,000 to 15,000 pounds per square inch (psi) for 15 seconds, minimum, without deformation or mechanical failure (see 6.4.1).

3.5 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 20 ± 1 pounds is applied at the steady rate of 0.125 inch per minute.

3.6 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled handtight.

3.7 Protective finish. Protective finish shall be in accordance with the applicable specifications and shall apply to all surfaces of the component regardless of contour configuration.

3.8 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using procedures in 4.6.5.

3.9 Workmanship. Initiators shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.9.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.9.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.9.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for

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acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: room ambient (68 to 77•F)
b. Altitude: sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psi

4.4 First article inspection. At the beginning of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of five metal parts assemblies, nine sets of unassembled parts, one set of unassembled parts without plating or final protective finish, and fifty shear pins from the same lot of shear pins tested by the contractor. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the first article sample shall be inspected for all requirements of the drawings and specifications. The Government reserves the right to require new first article samples until an acceptable sample is submitted (see 6.3).

4.5 Quality conformance inspection.

4.5.1 Lot.

4.5.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.5.1.2 Size of lot. The production lot size shall be as specified in the contract.

4.5.2 Examination. One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4x) magnification may be used in performing visual examination for

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such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.5.2.1 Chamber, Initiator, Drawing 10520407, covering a detail of Drawing 10520621.

Characteristics	Defect	Method of Inspection
Critical		
None defined		
Major		
101	Pitch diameter of large external thread, min	Gage
102	Major diameter of large external thread, min	Gage
103	Pitch diameter of small external thread, min	Gage
104	Major diameter of small external thread, min	Gage
105	Pitch diameter of internal thread, max	Gage
106	Minor diameter of internal thread, max	Gage
107	Concentricity of datum location with pitch diameter of small external thread	Gage
108	Concentricity of pitch diameter of large external thread with cavity	Gage
109	Concentricity of large external thread pilot with large external thread	Gage
110	Concentricity of counterbore cavity, including perpendicularity with base of counterbore	Gage
111	Diameter of counterbore	Gage
112	Diameter of large thread pilot	Gage
113	Depth of cavity	Gage
114	Length to datum location	Gage
115	Thickness of wall, min	Gage
116	Metal defective	Visual
117	Protective finish inadequate or defective	Visual
118	Surface finish of cavity and angle of gas port, improper	Visual
119	Burr or sharp corner	Visual
120	Foreign matter	Visual
Minor		
201	Pitch diameter of large external thread, max	Gage
202	Pitch diameter of small external thread, max	Gage
203	Pitch diameter of internal thread, min	Gage
204	Diameter of small external thread relief	Gage
205	Diameter of small external thread pilot	Gage
206	Diameter of hole in gas port	Gage
207	Diameter of cavity	Gage

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208	Depth of counterbore	Gage
209	Length of gas port	Gage
210	Length of small thread relief	Gage
211	Length from face of gas port to small thread	Gage
212	Length of large external thread pilot	Gage
213	Thickness of wall, max	Gage
214	Depth of threaded hole	Measure
215	Diameter of chamfer at internal thread	Measure
216	Total length	Measure
217	Length of large external thread	Measure
218	Marking missing, incorrect, or illegible	Visual
219	Radius or chamfer missing or incorrect	Visual
220	Surface finish improper, except as otherwise classified	Visual

4.5.2.2 Cap, Initiator, Drawing 10520473, covering a detail of Drawing 10520621.

Critical		
1	Surface finish of cavity	Gage
Major		
101	Pitch diameter of large internal thread, max	Gage
102	Minor diameter of large internal thread, max	Gage
103	Pitch diameter of small internal thread, max	Gage
104	Minor diameter of small internal thread, max	Gage
105	Pitch diameter of external thread, min	Gage
106	Major diameter of external thread, min	Gage
107	Pitch diameter of pin hole thread, max	Gage
108	Minor diameter of pin hole thread, max	Gage
109	Concentricity of striker hole with small cavity	Gage
110	Concentricity of large internal thread undercut with large thread	Gage
111	Concentricity of cartridge head recess with large internal thread	Gage
112	Concentricity of large internal thread with small cavity	Gage
113	Concentricity of chamfer at striker hole with small cavity	Gage
114	Concentricity of small internal thread with small cavity	Gage
115	Symmetry of pin hole with small cavity	Gage
116	Symmetry of flats with pitch diameter of thread at small end	Gage
117	Diameter of small cavity	Gage
118	Diameter of large thread undercut	Gage
119	Diameter of boss, min	Gage
120	Diameter of striker hole, min	Gage
121	Diameter of pin hole, max	Gage
122	Diameter of cartridge head recess	Gage
123	Diameter of counterbore at small thread cavity	Gage

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124	Diameter of mouth at small thread cavity	Gage
125	Depth of cartridge head recess	Gage
126	Depth to bottom of counterbore, small threaded end, max	Gage
127	Depth of cartridge head recess, including chamfer	Gage
128	Length of effective exterior thread, min	Gage
129	Length of end of effective exterior thread, max	Gage
130	Length of effective small interior thread, min	Gage
131	Thickness of boss, min	Gage
132	Location of pin hole	Gage
133	Chamfer at striker hole missing or incorrect	Visual
134	Metal defective	Visual
135	Protective finish inadequate or defective	Visual
136	Surface finish of cavity counterbore and mouth improper	Visual
137	Burr or sharp corner	Visual
138	Foreign matter	Visual

Minor

201	Pitch diameter of large internal thread, min	Gage
202	Pitch diameter of small internal thread, min	Gage
203	Pitch diameter of external thread, max	Gage
204	Pitch diameter of pin hole thread, min	Gage
205	Pitch diameter of setscrew thread	Gage
206	Minor diameter of setscrew thread, max	Gage
207	Diameter of boss, max	Gage
208	Diameter of striker hole, max	Gage
209	Diameter of pin hole, min	Gage
210	Diameter of body, large end	Gage
211	Depth to bottom of counterbore, small threaded end, min	Gage
212	Depth of small threaded cavity, including chamfer, max	Gage
213	Depth of large cavity, including undercut	Gage
214	Total length	Gage
215	Length of body, large end	Gage
216	Length from face of large cavity to shoulder	Gage
217	Length of flat	Gage
218	Width across flat	Gage
219	Width of large thread undercut	Gage
220	Thickness of boss, max	Gage
221	Location of setscrew hole	Visual
222	Radius or chamfer missing or incorrect	Visual
223	Surface finish improper, except as otherwise classified	Visual

4.5.2.3 Pin, Shear, Drawing 10520622, covering a detail of Drawing 10520621.

Critical
None defined

Major

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101	Diameter, max	Gage
102	Total length, min	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Surface finish improper	Visual
106	Burr	Visual
107	Foreign matter	Visual
Minor		
201	Total length, max	Gage
202	Profile of end	Measure

4.5.2.4 Filter, Drawing 10520414, covering a detail of Drawing 10520621.

Critical
None defined

Major		
101	Pitch diameter of thread, min	Gage
102	Major diameter of thread, min	Gage
103	Concentricity of thread with center hole and outside diameter	Gage
104	Symmetry of holes with outside diameter	Gage
105	Diameter of center holes	Gage
106	Diameter of holes	Gage
107	Depth of center hole	Gage
108	Depth of slot, max	Gage
109	Location of holes	Gage
110	Angular location of holes	Gage
111	Nylon patch not properly located or fused	Manual/Visual
112	Number of holes incorrect	Visual
113	Metal defective	Visual
114	Protective finish inadequate or defective	Visual
115	Burr	Visual
116	Foreign matter	Visual
Minor		
201	Pitch diameter of thread, max	Gage
202	Diameter of undercut	Gage
203	Diameter of body	Gage
204	Depth of slot, min	Gage
205	Total length	Gage
206	Length of body	Gage
207	Width of undercut, max	Gage
208	Width of slot	Gage
209	Location of slot	Gage
210	Marking missing, incorrect, or illegible	Visual

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211	Chamfer missing or improper	Visual
212	Surface finish improper	Visual

4.5.2.5 Pin, Firing, Drawing 10520474, covering a detail of Drawing 10520621.

Critical
None defined

Major

101	Concentricity of striker with body	Gage
102	Concentricity of groove with body	Gage
103	Symmetry of pin hole with body	Gage
104	Diameter of striker	Gage
105	Diameter of body	Gage
106	Diameter of groove	Gage
107	Diameter of pin hole	Gage
108	Depth of pin hole	Gage
109	Length of striker, min	Gage
110	Location of pin hole	Gage
111	Metal defective	Visual
112	Protective finish inadequate or defective	Visual
113	Surface finish of body improper	Visual
114	Burr or sharp corner	Visual
115	Foreign matter	Visual

Minor

201	Total length	Gage
202	Length of striker, max	Gage
203	Location of groove	Gage
204	Depth of slot	Measure
205	Width of slot	Measure
206	Width of groove	Measure
207	Location of slot	Measure
208	Angle on striker end of body	Visual
209	Radius or chamfer missing or incorrect	Visual
210	Surface finish improper, except as otherwise classified	Visual

4.5.2.6 Assembly, Drawing 10520621, less all Drawings except 10520473, 10520474, and MS29513-9.

Critical

1	Firing pin protrusion	Gage
2	O-ring missing	Visual

Major

101	Burr	Visual

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102	Foreign matter	Visual

Minor
None defined

4.5.2.7 Assembly, Drawing 10520621.

Critical
None defined

Major

101	Any part missing, except as otherwise classified	Visual
102	Improper assembly	Visual
103	O-ring not lubricated	Visual
104	Marking missing, incorrect, or illegible	Visual

Minor

201	Protective cap or plug missing	Visual
202	Protective finish inadequate or defective	Visual

4.5.2.8 Box, Fiberboard (Inner), Packing, Ammunition, Drawing 8830994.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual
202	Box damaged	Visual
203	Marking missing, incorrect, or illegible	Visual
204	Cut, damaged, or improperly sealed bag	Visual

4.5.2.9 Box, Packing, Ammunition, Drawing 8830995.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Visual/Manual

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202	Strapping missing, loose, broken, or improperly assembled	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.5.2.10 Box, Fiberboard (Outer), Alternative, Packing, Ammunition, Drawing 9217624.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual/Visual
202	Carton broken or damaged	Visual
203	Number of units contained incorrect	Visual
204	Marking missing, incorrect, or illegible	Visual
205	Tape incomplete	Visual

4.5.3 Inspection testing. The following tests shall be performed in accordance with 4.6.

4.5.3.1 Hardness. Fifty units from each heat treated lot of components shall be tested for compliance with the requirements on the applicable drawing. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot of components.

4.5.3.2 Hydrostatic pressure. Each chamber and cap shall be subjected to the hydrostatic pressure test. Failure of any cap or chamber subassembly to comply with 3.4 shall be cause for rejection of the subassembly.

4.5.3.3 Shear pin strength. Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with 3.5 shall be cause for rejection of the lot.

4.5.3.4 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.5.3.5 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.6.5. Failure of any unit in the inspection sample to comply with 3.8 shall be cause for rejection of the assembly.

4.6 Test methods and procedures.

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4.6.1 Hardness. Prior to the application of protective finish, a hardness tester shall be used to determine compliance with the requirements of the applicable drawing.

4.6.2 Hydrostatic pressure. Prior to application of protective finish the cap and chamber shall be mounted in the device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with 3.4.

4.6.3 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with 3.5. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.6.4 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.6.5 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (cap end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.7 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 8830994 and 8830995. Level B shall be in accordance with Drawings 8830994 and 9217624.

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-731-9272.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The initiator assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

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a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. If first article inspection is required (see 3.1 and 6.3)

d. The responsible inspection activity (see 4.1)

e. Any special instruction requirements (see 4.3)

f. Level of packing and marking (see 5.1)

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors that the items should be a first article sample. The first article should consist of five metal parts assemblies, nine sets of unassembled parts, one set of unassembled parts without plating or final protective finish, and fifty shear pins. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definition.

6.4.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.5 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.6 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

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Custodian:

Preparing activity
Navy - OS

Navy - OS

Air Force - 99

(Project 1377-0E01)

Review activity:
Air Force - 70

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